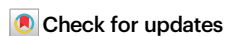


A liquid metal dynamic wetting strategy for spatiotemporal monitoring of hand movements

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Current hand tracking technologies often suffer from inconsistent accuracy due to occlusion, electromagnetic interference, and ambiguous signals from soft sensors. These limitations hinder reliable motion capture for advanced human-machine interaction. To overcome them, we propose a liquid metal dynamic wetting-based flexible and wearable inertial skeletal tracking (FWIST) system. The system integrates 16 FWIST units on the hand to capture palm pitch/roll and finger bending angles in real time with high precision. It accurately reconstructs fingertip and joint trajectories, providing comprehensive hand motion analysis. In laboratory tests, a feedforward neural network achieved nearly 100% gesture recognition accuracy on a small dataset, proving the method's potential. Future work will focus on verifying robustness on larger, diverse datasets. The FWIST approach offers promising applications in teleoperation, robot programming, drone control, and immersive VR/AR experiences.

The flexibility and multifunctionality of the human hand are distinguishing features that set it apart from the hands of animals, enabling it to perform a wide range of tasks, from simple grasping to complex sign language communication^{1–4}. As a result, real-time tracking and reconstruction of hand movements have long been a popular research topic and have found extensive applications in fields such as robotics⁵, healthcare⁶, gaming⁷, as well as virtual and augmented reality⁸. Researchers have conducted extensive studies on hand pose tracking and reconstruction^{9–13}. Nevertheless, the inherently complex structure of the hand, its numerous degrees of freedom, and the considerable anatomical variations among individuals present significant challenges for accurate hand motion estimation. Consequently, the majority of existing studies have been confined to limited scenarios and have not been able to fully capture and assess the entire spectrum of hand movements.

To address the issue of hand tracking, three primary approaches have been proposed. Firstly, vision-based hand tracking utilizes

cameras to achieve markerless hand motion tracking^{14–21}, often relying on machine learning techniques trained on large image datasets for processing^{22–24}. While machine learning demonstrates strong generalization in handling unseen hand shapes, object interactions, and varying lighting conditions, vision-based methods remain fundamentally constrained by occlusion issues²⁵. Secondly, IMU-/compass-based wearable hand tracking systems typically use six-axis IMUs (combining accelerometers and gyroscopes) and magnetic compasses, which are mounted on each bone of the hand to measure three-axis orientation data^{26–30}. When combined with additional hand position sensors, these systems can reconstruct hand movement^{31–37}. However, this approach has inherent limitations, such as susceptibility to magnetic field variations or interference. Furthermore, these sensing units are often rigid and bulky, negatively impacting wearability and user comfort, thereby limiting their practicality and user adoption^{38,39}. Finally, flexible wearable hand tracking systems leverage multiple flexible sensors to

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estimate hand postures, generating signals in response to deformation. These sensors are attached to the hand and, when combined with additional posture sensing technologies, can optimize hand tracking^{40–45}. However, these systems also have inherent drawbacks. Firstly, they struggle to distinguish between deformations caused by hand movements and those resulting from contact with objects. As a result, their performance becomes inadequate for scenarios where users manipulate objects, use tools, or wear tactile devices. Additionally, calibration challenges arise due to sensor hysteresis and coupling effects between sensors. Their durability is also limited, as the sensors are prone to damage under large deformations or repeated bending. These issues significantly hinder the reliability and practicality of this approach in real-world applications.

Enhancing methodologies for hand motion analysis is crucial to achieving more comprehensive and adaptable results. To tackle existing limitations, we propose a liquid metal (LM) dynamic wetting triggered FWIST sensing strategy. Our approach involves creating a flexible, wearable, miniature FWIST unit by injecting LM, which possesses excellent conductivity and fluidic properties, into microchannels with scale-like microstructured inner walls. These specially structured microchannels are achieved by replicating the morphology of horsehair surfaces (Supplementary Figs. 1–5). Significantly, unlike traditional wearable sensors that are typically placed at joint locations, our FWIST unit is strategically mounted at bone positions (Fig. 1a). This innovative design results in a solid (PDMS, polydimethylsiloxane)-liquid (LM)-gas (air) three-phase interface, characterized by a pressure gradient. The dynamic additional pressure associated with the tilt angle induces a shift in the gas-liquid interface, which in turn alters the surface morphology of the LM. This change enables the acquisition of tilt angle data through precise resistance measurements. (Fig. 1b). In practical applications, the FWIST on the surface of the bone can provide pitch and roll angle information, which can be used to reconstruct the hand model during the processes of pitching, rolling, and grasping (Fig. 1a). This proposed hand tracking strategy offers accurate, robust, and cost-effective spatiotemporal monitoring of hand motions, addressing the fundamental limitations of existing methods. Key advantages of our FWIST sensing strategy include its independence from visual occlusions and complex environments; the system's functionality is unaffected by these challenges. Additionally, the FWIST unit is designed for wearability, being fully flexible with a compact volume of 1 mm³. It provides comfort and ease of wear without deformation during hand movements.

Furthermore, the system's monitoring range and utility can be significantly enhanced through the integration of machine learning techniques, thereby increasing its versatility and potential applications (Fig. 1c). For instance, in medical rehabilitation, this approach can play a crucial role in enabling communication for individuals with speech impairments by detecting their hand movements and converting these gestures into speech. In the realm of human-computer interaction, integrating this technology directly into wearable interfaces has the potential to significantly enhance precision control capabilities within virtual reality/augmented reality systems. Furthermore, in the domain of flexible robotics, the sensing units can be strategically embedded within robotic systems in a “vascular-like” configuration, effectively functioning as both conductive pathways and posture sensors.

Results

The working mechanism of FWIST unit

We precisely controlled the pitch angle (θ) and roll angle (ω) of the FWIST unit using a dual-axis motorized rotary stage (Fig. 2a). Simultaneously, the resistance signal characteristics of a sample with a length of L and an inner diameter of D were collected under different θ and ω values using an Agilent 34420 A (Fig. 2b). At this point, at a distance s from the origin, the LM deforms under the additional pressure at the three-phase interface on the inner wall of the

microchannel. The variation in additional pressure P is mainly influenced by θ and ω , and can be expressed as:

$$P(s, \omega, \theta) = \rho g s \cos(\omega) \sin(\theta) \quad (1)$$

where the g is a constant value^{46–50} (Supplementary Figs. 6–8). When the FWIST unit is placed on the motorized rotary stage at a fixed ω , the additional pressure P at position s gradually increases, and the larger the s value, the greater the change in additional pressure, that is, $P(x, \omega, \theta_0) < P(s, \omega, \theta_1) < P(s, \omega, \theta_2)$. Consequently, the liquid surface of the LM becomes smoother, and the resistance signal is expected to exhibit a cosine function characteristic as θ continues to increase (Fig. 2c). We investigated the resistance response characteristics of FWIST unit as a function of θ under varying ω values. The experimental data reveal a strong correlation between the resistance signal changes and the parameters θ and ω . Specifically, as θ increases from -90° (vertical state) to 270° , the resistance change rate exhibits a sinusoidal trend. Additionally, as ω increases, the amplitude of the resistance changes rate decreases progressively, following a cosine-like relationship with respect to ω (Fig. 2d and Supplementary Video 1).

Another significant factor contributing to the observed strong correlation between the resistance signal changes and the parameters θ and ω is the tight adhesion between the LM and the PDMS substrate (Supplementary Fig. 9). This adhesion prevents the sliding of the three-phase interface during deformation. Moreover, the adhesion of the oxide layer significantly alters the wettability of the PDMS surface with respect to the LM, which in turn influences the migration dynamics of the gas-liquid interface. To explore this effect, we measured the contact angle of LM on two distinct surfaces, which are untreated PDMS and PDMS surfaces coated with the LM's native oxide layer. On pristine PDMS, the apparent contact angle was 150.8° , indicating a strong non-wetting behavior. In contrast, the contact angle on the oxide-coated PDMS was dramatically reduced to 16.8° , reflecting a significant enhancement in wettability (Supplementary Fig. 10). These findings underscore that the oxide layer plays a critical role in modifying the interaction between PDMS and the LM, thereby influencing the migration of the gas-liquid interface, particularly at the boundary between untreated PDMS regions and those with the LM oxide coating. Specifically, the Raman spectrum of PDMS indicates the presence of Si-O-Si bonds and $-\text{CH}_3$ groups on its surface (Fig. 2e and Supplementary Fig. 11). Meanwhile, X-ray photoelectron spectroscopy (XPS) confirms the presence of an oxide layer on the surface of LM. The Ga 3d spectrum establishes the surface oxidation state of LM, where exposure to air leads to the formation of a Ga_2O_3 layer corresponding to the Ga (III) peaks, with a binding energy of 20.4 eV (Fig. 2f and Supplementary Fig. 12). This suggests that the adhesion between PDMS and LM primarily arises from van der Waals forces between Ga and O atoms on the LM surface and Si, O, C, and H atoms on the PDMS surface. Although these forces are fundamentally weak interactions, the oxide layer on the LM acts as a “soft solid,” allowing extensive conformal contact with the PDMS surface. Consequently, despite the weak interaction among individual atoms, the large overall contact area results in a cumulative effect that renders the van der Waals forces significantly impactful at the macroscopic level, thereby facilitating effective adhesion (Fig. 2g).

To further investigate the microchannel phenomena that are challenging to characterize and observe directly, we conducted simulations and modeling of the dynamic deformation behavior of the LM under the influence of additional pressure within the microchannel. The simulation results demonstrate that, upon injection into the microchannel, the LM does not fully occupy the microstructural valleys, instead forming a clear gradient distribution. Specifically, the metal content decreases progressively from the inlet (left) to the outlet (right) (Fig. 2h). Furthermore, during the rotation of the FWIST unit, the LM content within the microstructural valleys dynamically changes

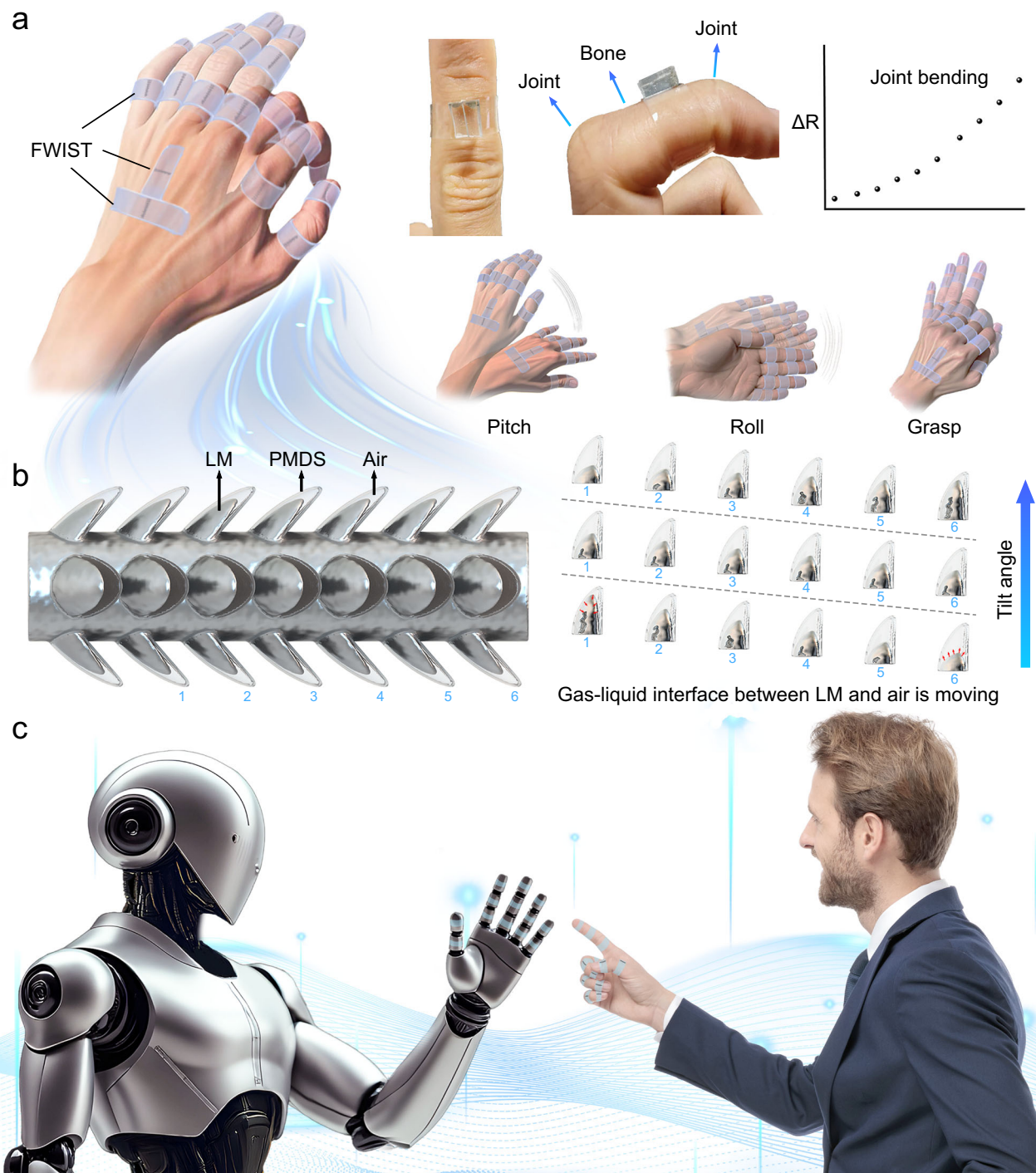


Fig. 1 | Design strategy and potential applications of FWIST. a Wearing method of FWIST. **b** Schematic diagram of the gradient distribution of LM within the microchannel and the movement of the gas-liquid interface during the tilt angle

change of the FWIST unit. **c** Potential applications of FWIST in the fields of medical rehabilitation and virtual reality/augmented reality.

in response to variations in the angular position, θ , driven by changes in additional pressure. For example, the LM content in the microstructural valley outlined by the black box varies as θ progresses from 0° to 90° , 180° , and 270° . The corresponding LM content changes from 77.51% to 77.35%, 77.82%, and 77.52%, respectively. Similarly, the LM content at different positions within the microstructural valleys also changes as θ increases. Notably, the variation in LM content follows a sine-function-related trend with respect to θ (Fig. 2i). In addition, we simulated the resistance response characteristics of FWIST

unit under varying angular velocities to examine their dependence on both θ and ω . The simulation results reveal a strong and quantifiable correlation between these two parameters and the resistance signal changes of FWIST unit. Specifically, as θ increases from -90° (vertical state) to 270° , the resistance changes rate exhibits a sinusoidal trend. Moreover, as ω increases, the amplitude of the resistance change rate gradually diminishes, demonstrating a cosine-like trend in relation to ω (Fig. 2j and Supplementary Fig. 13). These findings are in strong agreement with experimental observations, further validating the

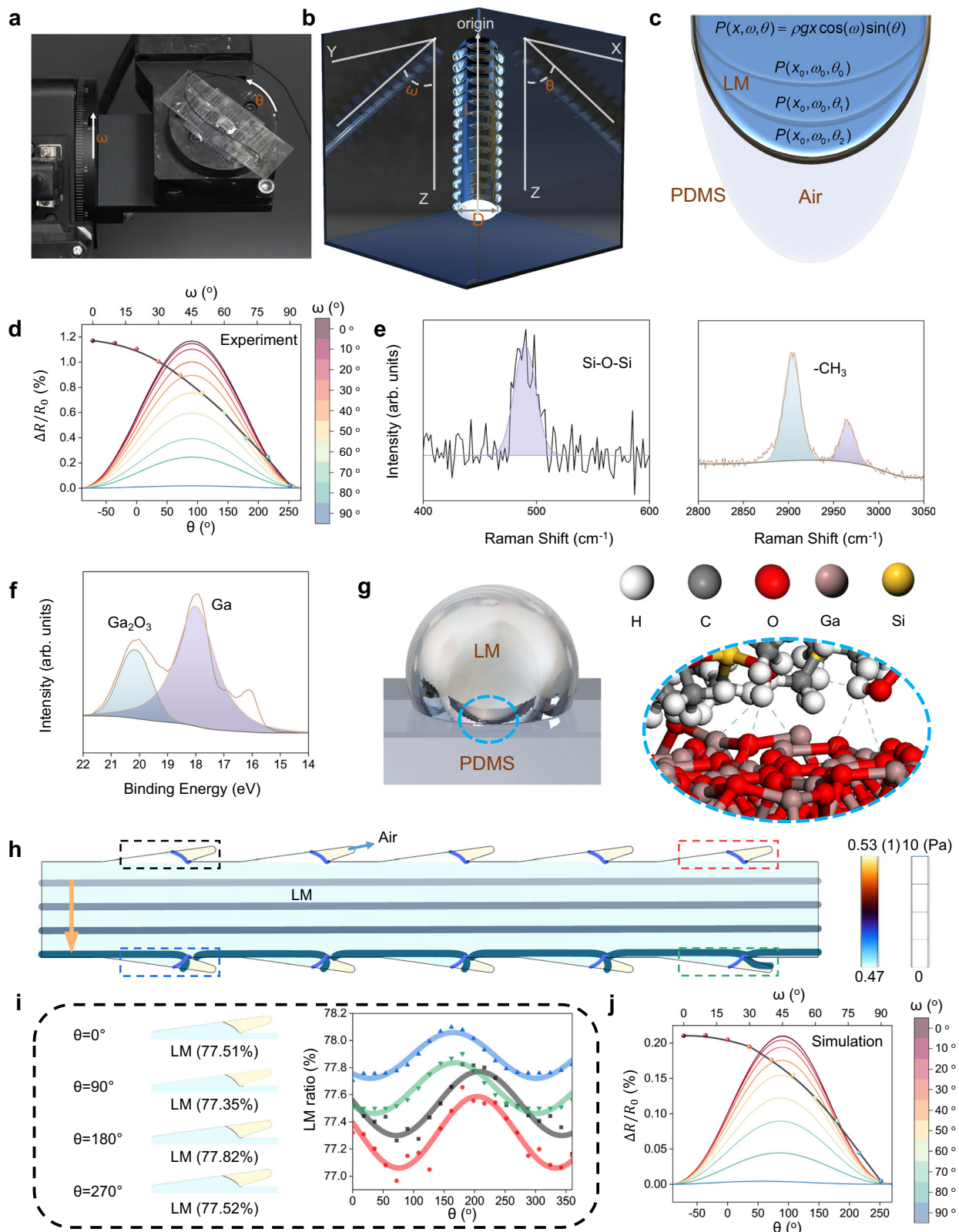


Fig. 2 | The working mechanism of FWIST unit. **a** FWIST unit testing setup using a motorized rotary stage. **b** Parameters employed during FWIST testing. **c** Deformation of LM at the three-phase interface within the microstructural valley of the microchannel under additional pressure. **d** Electrical signal response characteristics of FWIST as a function of θ and ω . **e** Raman spectroscopy analysis of the PDMS surface, indicating the contribution of Si-O-Si and -CH₃ groups. **f** XPS analysis

of the LM surface, highlighting the presence of Ga₂O₃. **g** Schematic diagram of the adhesion between PDMS and LM. **h** Simulation of LM distribution within the microchannel after injection. **i** Simulation of LM content variation (deformation) within the microstructural valley during FWIST rotation. **j** Simulation of FWIST's electrical signal response as a function of θ and ω .

robustness of our simulations and the accuracy of the experimental setup.

The working characteristics of FWIST unit

The FWIST unit was mounted on the axle head of a spool to examine its operational mechanism by monitoring the rotational motion of the spool's axis during the unrolling process. As the spool unfolds, the FWIST unit undergoes periodic rotational motion that is precisely synchronized with the axis's unrolling. This rotational movement induces corresponding periodic variations in the electrical signal generated by the unit. Notably, these electrical signal variations exhibit a distinct sinusoidal pattern, which reflects the intrinsic coupling between the rotational dynamics of the FWIST unit and its electrical response. This behavior provides a clear demonstration of the FWIST unit's functionality and its sensitivity to periodic motion, showcasing its potential utility in dynamic sensing applications (Fig. 3a). When the FWIST unit is initially set to a value of -90° and continuously rotates counterclockwise through a full 360° cycle, the generated electrical signal exhibits a well-defined sinusoidal variation (Fig. 3b). Within the operational range of -90° to 90° , the FWIST unit achieves a high resolution of 2° , and the corresponding signal variations are characterized by exceptional accuracy, stability, and the absence of drift. This consistent performance highlights the reliability of the FWIST unit for precise angular measurements under static conditions.

Beyond its excellent static testing capabilities, the FWIST unit also demonstrates remarkable dynamic detection performance, making it suitable for applications involving rapid angular transitions. When subjected to fast angular changes, the FWIST unit responds with high efficiency, being capable of generating a step response within just 0.27 s (Fig. 3c and Supplementary Video 2). This rapid response is indicative of the device's superior sensitivity and robustness, underscoring its potential for use in dynamic sensing scenarios where speed and precision are critical. We further investigated the lag characteristics of the electrical signal variations of the FWIST unit during continuous rotation, which arise due to the inherent response time of the device. As the rotational frequency of the FWIST unit increased from 0.034 Hz to 0.34 Hz, the amplitude of the electrical signal variation within a single rotational cycle decreased from 1.17% to 1.1% (Fig. 3d and Supplementary Fig. 14). This reduction in signal amplitude with increasing rotational frequency highlights the influence of the response time on the device's performance at higher operating speeds. These findings provide insights into the dynamic behavior of the FWIST unit and suggest areas for potential optimization to maintain signal fidelity under rapid rotational conditions.

In contrast to traditional flexible sensors that rely on large-scale deformations, the resistance variation of our FWIST unit is primarily driven by the localized, small-scale deformation of the LM within the anisotropic microchannel structures. These controlled deformations prevent the rupture of the LM's oxide layer, thereby ensuring signal stability during operation. Furthermore, the cumulative effects of the numerous microstructures amplify the signal magnitude without introducing drift, as demonstrated by the sensor's stable performance over more than 5000 cycles of testing (Fig. 3e and Supplementary Fig. 15). In addition to its excellent cyclic stability, the FWIST unit exhibits remarkable long-term reliability, maintaining consistent performance over a duration of 30 days (Fig. 3f). These findings underscore the FWIST unit's robustness and highlight its potential for applications requiring high durability and stability under both dynamic and long-term operational conditions.

Furthermore, we performed a systematic analysis of the factors influencing the amplitude of electrical signal variations, specifically focusing on the LM injection speed during sample preparation, as well as the rotational frequency and ω of the sample during operation. The experimental results reveal that both the injection speed during fabrication and the ω have a pronounced effect on the signal amplitude.

Importantly, the observed trends in the experimental data closely align with the results of simulation studies, thereby validating the experimental findings and confirming the robustness of the predictive model (Fig. 3g, h and Supplementary Figs. 16–20). This alignment between experimental and simulation data underscores the critical role of precise control over both fabrication parameters and operational conditions in optimizing the performance of the FWIST unit. By understanding and fine-tuning these factors, it is possible to enhance the device's performance while ensuring consistent and reliable signal output across a range of applications.

FWIST for monitoring hand movements

In daily life, the range of motion of the human hand primarily consists of three key types of movements: the pitch movement of the palm (upward and downward tilting motion), the rotational movement of the palm (pronation and supination), and the grasping or flexion-extension movements of the fingers. These motions are fundamental to performing various tasks, from simple daily activities to complex manipulations, and highlight the versatility and functional significance of the human hand in adaptive interactions with the surrounding environment. When FWIST is applied to track hand posture, a total of 16 FWIST units are required to work in coordination. Among them, two FWIST units (unit-1 and unit-2) are configured vertically on the back of the hand to track overall hand movement, while the remaining 14 units (unit-3 to unit-16) are distributed along the phalanges of each finger, oriented axially along the length of the fingers. Specifically, FWIST unit-1 is responsible for monitoring the overall pitch angle (θ) of the hand, while FWIST unit-2 is used to track the overall roll angle (ω) of the hand. The other 14 FWIST units are utilized to monitor the bending angles (θ) of the individual phalanges, providing detailed information on the flexion and extension of each finger (Fig. 4a and Supplementary Fig. 21). And by explicitly defining the direction of the arrow to represent the injection direction of the LM within the channel, thus clarifying the orientation of the FWIST on the finger. Specifically, when the arrow points vertically downward, we define the tilt angle as 0° , when the arrow points vertically upward, the tilt angle is defined as 180° . This configuration enables comprehensive tracking of hand pose and motion, providing high-resolution data for applications such as gesture recognition, biomechanical analysis, and human-machine interaction. It also offers robustness against sweat, dirt, and grime, ambient temperature variations, and repeated mounting and removal (Supplementary Figs. 22–24).

The range of pitch motion for the human palm typically spans from -90° to 90° . Within this range, there exists a clear one-to-one correspondence between the resistance signal variation rate of FWIST unit-1 and the tilt angle, allowing for accurate identification of the palm's pitch posture using FWIST unit-1 alone (Fig. 4b). Similarly, the range of roll motion for the human palm is -90° to 180° . In the range of -90° to 90° , the electrical signal variation rate of FWIST unit-2 also exhibits a one-to-one correspondence with the rotational angle, enabling precise monitoring of the palm's roll angle. However, within the range of 90° to 180° , the electrical signal variation overlaps with that observed in the 0° to 90° range, making it challenging to unambiguously determine the roll angle using FWIST unit-2 alone (Fig. 4c). To address this limitation, FWIST unit-8 can be used to assist in achieving precise identification of the roll angle. Specifically, in the rotational range of 0° to 90° , the electrical signal of FWIST unit-8 exhibits an increasing trend, whereas in the range of 90° to 180° , the electrical signal exhibits a decreasing trend. This behavior is influenced by the structural characteristics and bending direction of the human fingers, which affect the measurements from FWIST unit-8. Once the pitch and roll angles of the palm are determined through these combined measurements, it becomes straightforward to monitor the bending states of the fingers and further track the current spatial posture of the entire hand.

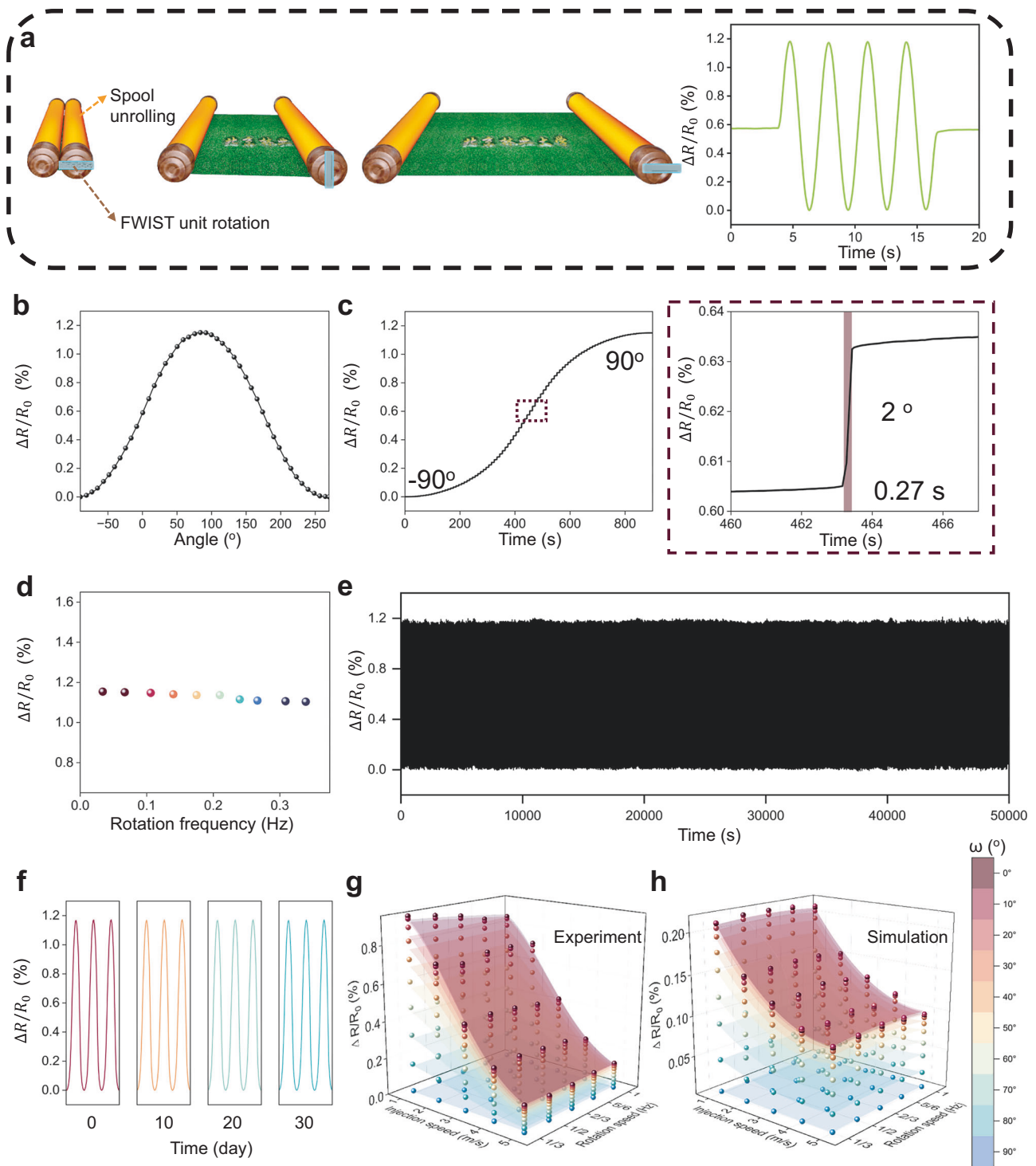
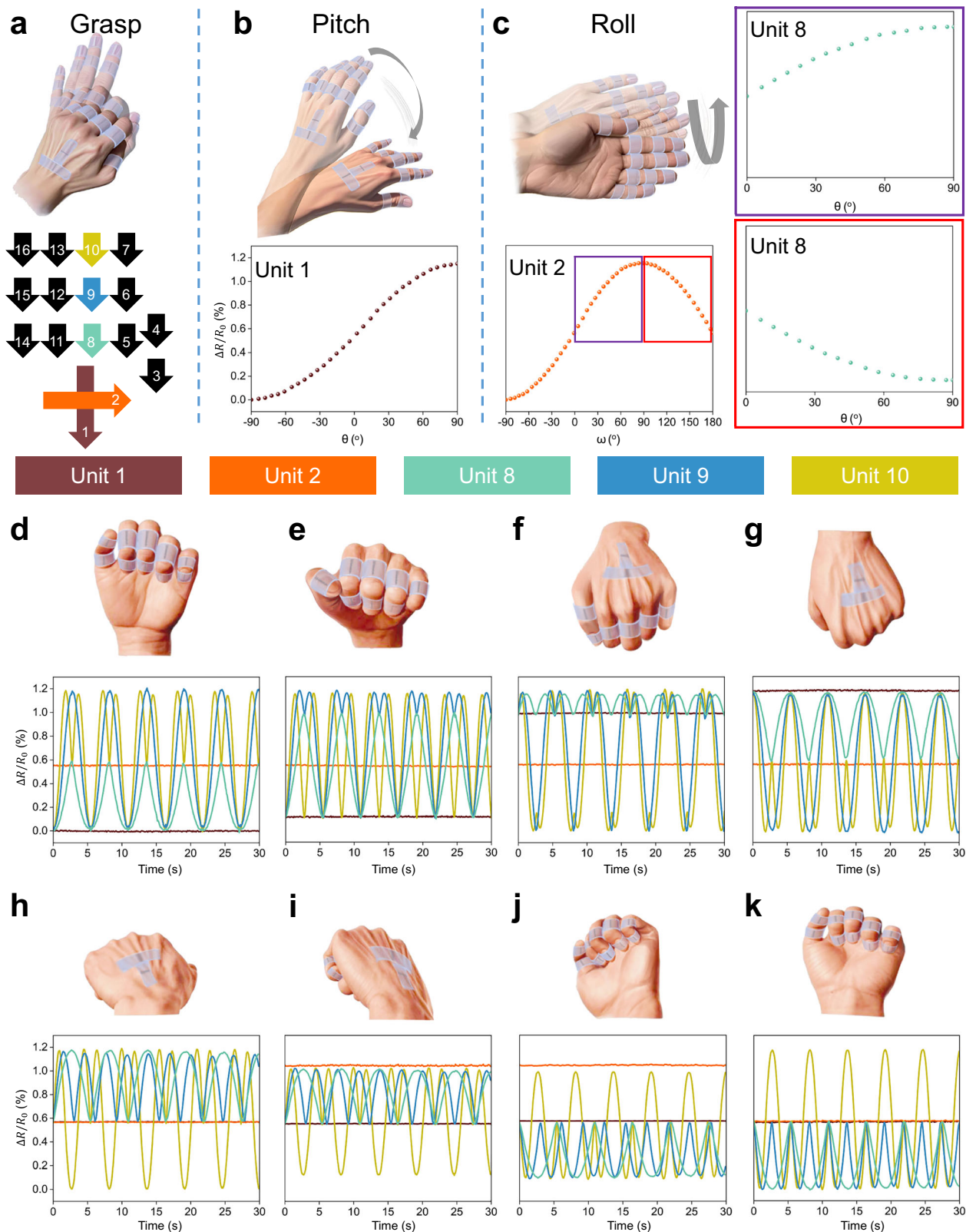


Fig. 3 | The working characteristics of FWIST unit. **a** Schematic diagram of the FWIST unit working process (Monitoring spool unrolling). **b** The relationship between θ ($\omega = 0^\circ$) and resistance variation rate. **c** The monitoring accuracy and response time of the FWIST unit. **d** The effect of rotational frequency on resistance signal variation rate. **e** Cyclic stability testing of the FWIST unit. **f** Long-term

durability testing of the FWIST unit. **g** The influence of injection speed, rotational frequency, and tilt angle (ω) on the electrical signal variation rate of the FWIST unit. **h** Simulation of the influence of injection speed, rotational frequency, and tilt angle (ω) on the electrical signal variation rate of the FWIST unit.

We illustrate this by tracking gripping actions when the palm is in eight different postures using FWIST units 1, 2, 8, 9, and 10 ($\omega = 0^\circ$, $\theta = 0^\circ$, 45° , 135° , 180° ; $\theta = 90^\circ$, $\omega = 0^\circ$, 45° , 135° , 180°). When $\omega = 0^\circ$ and $\theta = 0^\circ$, 45° , 135° , 180° , the roll and pitch angles exhibit a one-to-one correspondence with the rate of change in the electrical signal, highlighting a direct relationship that underscores their interdependence.

Consequently, the hand gesture posture can be easily identified (Fig. 4d–g). Similarly, when $\theta = 90^\circ$ and $\omega = 0^\circ$, 45° , 135° , 180° , the pitch angle of the hand posture can be directly determined using FWIST unit-1. While the electrical signal variation rates are identical at $\omega = 45^\circ$ and $\omega = 135^\circ$, FWIST unit-8 distinguishes between these by showing a gradually increasing electrical signal variation rate at $\omega = 45^\circ$ and a gradually



decreasing rate at $\omega = 135^\circ$ (Fig. 4h–k). These distinct trends indicate that precise hand posture recognition can be achieved by simultaneously utilizing multiple FWIST units, demonstrating the system’s capability for detailed spatial posture tracking. And this capability highlights the effectiveness of the FWIST system for accurately capturing complex hand postures, making it valuable for applications such as gesture recognition, biomechanics analysis, and human-computer interaction.

FWIST for spatiotemporal monitoring of hand movements

We equipped the hand with the FWIST system and sequentially executed actions 1 through 8 to thoroughly evaluate its performance. The system comprises 16 FWIST units working collaboratively to capture comprehensive posture information of the hand (Fig. 5a). Specifically, FWIST unit-1 is responsible for measuring the hand’s pitch angle, providing insights into its vertical orientation, while FWIST unit-2

Fig. 4 | FWIST monitors the grasping actions of the middle finger in different spatial positions of the hand. **a** Schematic diagram of FWIST assembly on the hand. **b** The rate of change in the electrical signal associated with hand pitching movements. **c** The rate of change in the electrical signal associated with hand rolling movements. And the rate of change of the electrical signal from FWIST unit-8 during hand grasping movements when x is in the ranges of $0-90^\circ$ and $90-180^\circ$. **d** The rate of change of the electrical signal corresponding to FWIST units 1, 2, 8, 9, and 10 during hand grasping movements when θ is -90° and $\omega = 90^\circ$. **e** The rate of change of the electrical signal corresponding to FWIST units 1, 2, 8, 9, and 10 during hand grasping movements when θ is -45° and $\omega = 90^\circ$. **f** The rate of change of the electrical signal corresponding to FWIST units 1, 2, 8, 9, and 10 during hand

grasping movements when θ is 45° and $\omega = 90^\circ$. **g** The rate of change of the electrical signal corresponding to FWIST units 1, 2, 8, 9, and 10 during hand grasping movements when θ is 90° and $\omega = 90^\circ$. **h** The rate of change of the electrical signal corresponding to FWIST units 1, 2, 8, 9, and 10 during hand grasping movements when θ is 0° and $\omega = 0^\circ$. **i** The rate of change of the electrical signal corresponding to FWIST units 1, 2, 8, 9, and 10 during hand grasping movements when θ is 0° and $\omega = 45^\circ$. **j** The rate of change of the electrical signal corresponding to FWIST units 1, 2, 8, 9, and 10 during hand grasping movements when θ is 0° and $\omega = 135^\circ$. **k** The rate of change of the electrical signal corresponding to FWIST units 1, 2, 8, 9, and 10 during hand grasping movements when θ is 0° and $\omega = 180^\circ$.

records the roll angle, essential for understanding lateral movements. The remaining units are meticulously dedicated to capturing the bending angles of each finger bone, offering detailed analysis of finger joint dynamics (Fig. 5b). This configuration enables precise monitoring and analysis of complex hand movements for various applications.

Furthermore, leveraging the capabilities of MATLAB, we developed a comprehensive 3D model of the hand (Fig. 5c). We meticulously imported the time-dependent angular data acquired by each FWIST unit into the MATLAB environment to accurately simulate and control the hand model's dynamic movements. The angular measurements from FWIST unit-1 and unit-2 were strategically utilized to govern the overall hand's pitch and roll angles, respectively, offering critical insight into the hand's orientation in three-dimensional space. In contrast, the remaining 14 FWIST units were exclusively tasked with regulating the precise bending angles of each individual finger bone, thus providing a detailed depiction of finger articulation. In our analysis, we paid particular attention to the intricate movement patterns of the middle finger, specifically tracking the trajectories of the joint-1, joint-2, and fingertip as the hand performed sequential actions 1 through 8 (Fig. 5d, e and Supplementary Video 3).

We generated projections of these trajectories on the X - Y , X - Z , and Y - Z planes to visualize and analyze the spatial dynamics. We observed that certain actions yielded identical motion paths for the middle finger, prompting us to selectively illustrate the trajectories for actions I, III, IV, V, and VI in the projection planes for clarity (Fig. 5f). The results of these trajectory analyses underscore the efficacy of our FWIST system, demonstrating its robust capability to capture and analyze the complex spatiotemporal characteristics of hand movements in a detailed and reliable manner. We also developed an advanced feedforward neural network designed with two hidden layers to analyze hand movement data. The architecture of the network includes 20 neurons in the first hidden layer and 15 neurons in the second hidden layer, optimized to capture the nuances of hand postures. We systematically collected electrical signals for hand actions I-VIII (50 trials per action), yielding a relatively small dataset of 400 samples with limited diversity. After random shuffling to mitigate bias, the data were split into training, validation, and test sets (70%, 15%, and 15%, respectively). On this dataset, our machine-learning model achieved 100% accuracy in classifying the eight hand postures, but these results remain preliminary and are confined to the current, small-scale data. While the findings suggest that our neural-network architecture can capture complex, multi-degree-of-freedom hand dynamics (Fig. 5g and Supplementary Figs. 25–28), further validation on larger and more varied datasets is required to confirm robustness and generalizability. These initial outcomes indicate that FWIST may serve as a valuable tool for precise hand-motion monitoring, with future work aimed at broader experimental evaluation and real-world application.

Discussion

This study proposes an innovative wearable FWIST system for spatiotemporal monitoring of hand movements. Unlike conventional vision or glove solutions, FWIST exploits the dynamic wetting

behavior of LM under applied pressure to transduce tilt angles directly into electrical signals. Finite-element simulations and experimental measurements consistently validate the reliability, repeatability, and precision of this mechanism. Each module maintains stable performance over 5000 continuous rotation cycles and operates reliably for at least 30 days. The system achieves an angular resolution of 2° and a response time of 0.27 s, enabling continuous, precise monitoring of subtle hand gestures. Its compact, conformal design adheres unobtrusively to the hand without impeding natural interactions, thereby overcoming the portability and interference limitations of existing technologies.

To assess the potential of FWIST in gesture classification, we trained a feedforward neural network on data from a homogeneous subject cohort and achieved 100 percent classification accuracy in preliminary experiments. However, these results are based on a limited, uniform dataset. A thorough evaluation of robustness and generalizability will require validation with larger, more diverse populations and in real-world scenarios. Although FWIST demonstrates considerable promise, it remains at the prototype stage. The current design requires 16 sensors per hand, which may hinder adoption in cost and space constrained applications. To improve manufacturability and reproducibility, we propose replacing horsehair-mold fabrication with precision-printed polymer masters, an approach that awaits experimental verification. Future work will focus on expanding and diversifying the training dataset, optimizing sensor integration, validating scalable manufacturing processes, and conducting in situ performance tests to support the utilization of FWIST in scientific and interdisciplinary research.

Methods

Preparation of FWIST unit

First, we utilized PDMS silicone to replicate the surface morphology of horsehair, maintaining a mass ratio of 10:1 between PDMS and the curing agent. After immersing the horsehair in PDMS, it was placed in a vacuum oven and subjected to -0.08 MPa for 0.5 h. The pressure was then restored to atmospheric levels, and the assembly was heated at 65°C for 2 h to ensure complete curing. Subsequently, the horsehair was removed, yielding microchannels with a scale-like inner wall morphology.

The FWIST unit was fabricated by directly injecting LM into these microchannels at a speed of 1 m s^{-1} , with the injection rate controlled by a peristaltic pump. Once the LM was injected into the microchannels, the ends of the channels were sealed with conductive silver paste, and wires were connected to form electrodes. Finally, the device was placed in a vacuum oven at 65°C for 1 h to sinter and cure the conductive silver paste.

Record of electrical signals

The electrical signals generated during the rotational movement of the sample were meticulously measured using a high-precision digital multimeter (Agilent 34420A, USA). Meanwhile, the data were collected using the accompanying software of the device, BenchVue Basic Digital Multimeter App (2019Update4.0).

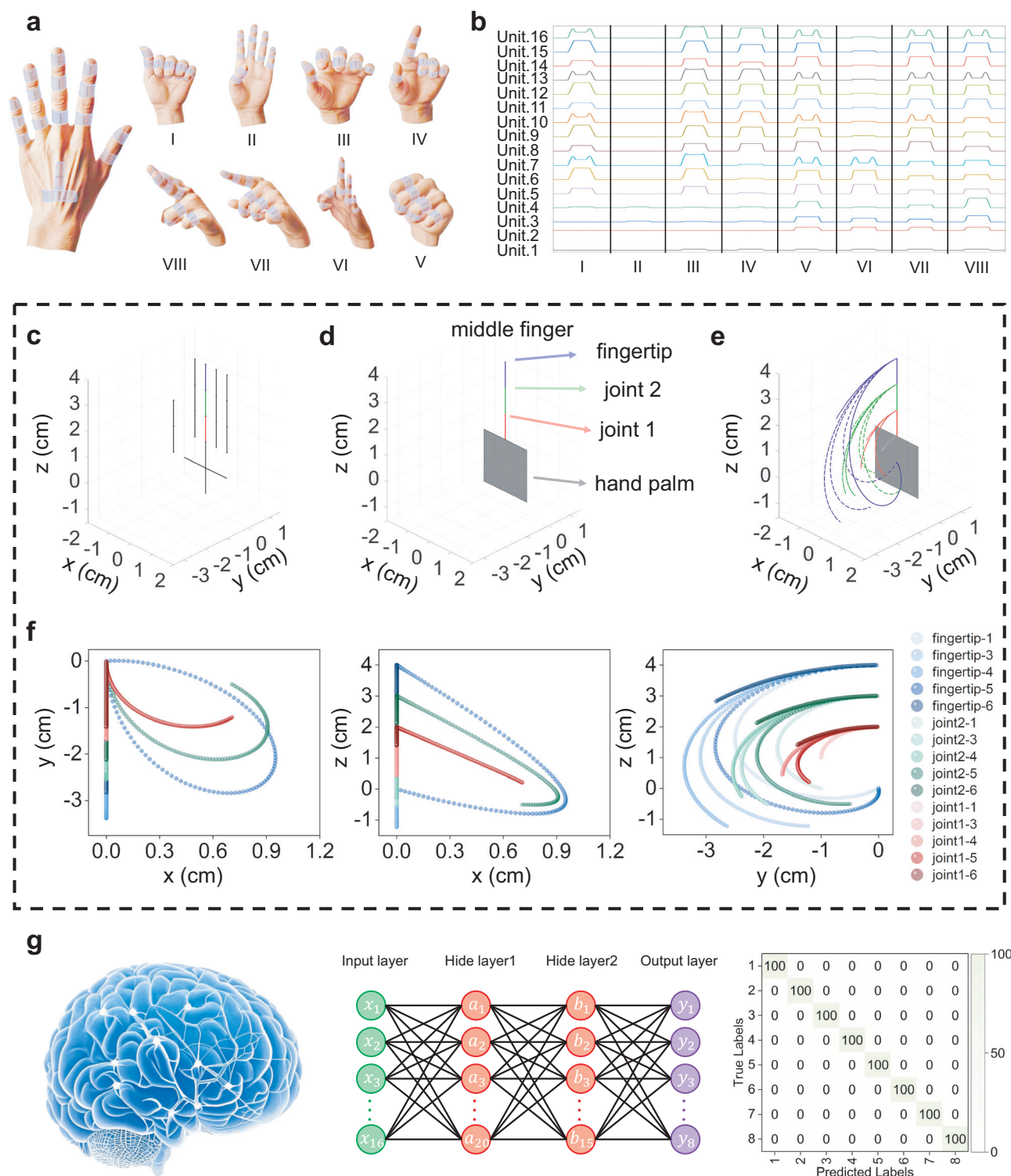


Fig. 5 | FWIST for spatiotemporal monitoring of hand movements. **a** Schematic of FWIST fitted on the hand and the 8 different hand postures. **b** The changes in electrical signals collected by 16 different FWIST units as the hand performs actions 1 through 8 sequentially. **c** Hand model established using MATLAB. **d** The position of the middle finger's fingertip, joint and palm. **e** The movement trajectory of the

middle finger's fingertip, joints, and palm during the movement process. **f** The projection of the movement trajectory on the X - Y , X - Z , and Y - Z planes. **g** The network diagram of the feedforward neural network model used for hand gesture recognition and the confusion matrix.

Raman analysis of PDMS

The Si-O-Si and -CH₃ groups on the surface of PDMS were determined by confocal Raman microscope (inVia-Qontor).

XPS analysis of LM

The Ga₂O₃ of LM surface was determined by X-Ray Photoelectron Spectroscopy (ESCALAB250 Xi).

SEM characterization

The sample was mounted on the stage with conductive adhesive and gold-sputtered for 60 s prior to testing. Imaging was performed on a Hitachi SU8600 field-emission scanning electron microscope in secondary-electron mode at an accelerating voltage of 3 kV.

Simulation analysis

Numerical theory simulations were performed using the commercial software COMSOL Multiphysics (Version 5.6, COMSOL Inc., Sweden). All computational models have been analyzed for mesh independence, and a medium number of meshes has been used after comprehensive consideration.

Statistics and reproducibility

No statistical method was used to predetermine the sample size. No data were excluded from the analyses. The experiments were not randomized. The investigators were not blinded to allocation during experiments and outcome assessment.

Ethics declaration

The study protocol was reviewed and approved by the Institutional Human Study Ethics Committee of Beijing Forestry University (BJFUPSY-2025-075), which is affiliated with Beijing Forestry University (Beijing, China) and conducted in full compliance with relevant ethical guidelines. All participants provided written informed consent before taking part in the experiments. Written consent for the publication of any images or data that could potentially identify participants, including the main figures, Supplementary Figs. and the schematics in our point-by-point response to reviewers were also obtained.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The authors declare that all data supporting the findings of this study are available within the article and its supplementary files. Any additional requests for information can be directed to and will be fulfilled by the corresponding authors. Source data are provided with this paper.

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Author contributions

L.W. and G.S. supervised and guided the experiment. L.W. and F.Z. conceptualized the research idea and designed the experiment. N.L. and R.Z. assisted with the experiment, contributed to theoretical analysis, and participated in simulation work. C.C., G.W., F.G., and B.T. took part in the experiment and data collection. H.Z. and S.W. contributed to the discussion. F.Z. wrote the manuscript, L.W. and G.S. revised it. All authors have read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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